In [2]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
%matplotlib inline
import seaborn as sns
```

In [3]:
```
#import csv file
df=pd.read_csv('Sales_Data.csv', encoding='unicode_escape')
```

In [9]:
```
#Check number of rows and columns
df.shape
```

Out[9]: (11257, 13)

In [10]:
```
#Too see the imported data
df.head(5)
```

Out[10]:

| | User_ID | Cust_name | Product_ID | Age | Age Group | Gender | State | Zone | Zi |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 1002903.0 | Anvi | P00125942 | 27.0 | 26-35 | Female | Maharashtra | West | |
| **1** | 1000732.0 | Shanta | P00110942 | 34.0 | 26-35 | Female | Andhra Pradesh | South | |
| **2** | 1001990.0 | Sheetal | P00118542 | 16.0 | 0-17 | Female | Uttar Pradesh | Central | |
| **3** | 1001425.0 | Virendra | P00237842 | 16.0 | 0-17 | M | Karnataka | South | |
| **4** | 1000588.0 | Vishal | P00057942 | 28.0 | 26-35 | M | Gujarat | West | |

In [11]:
```
#Field details and data type
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11257 entries, 0 to 11256
Data columns (total 13 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   User_ID           11245 non-null  float64
 1   Cust_name         11245 non-null  object
 2   Product_ID        11245 non-null  object
 3   Age               11245 non-null  float64
 4   Age Group         11245 non-null  object
 5   Gender            11245 non-null  object
 6   State             11245 non-null  object
 7   Zone              11245 non-null  object
 8   Zipcode           0 non-null      float64
 9   Profession        11245 non-null  object
 10  Product_Category  11245 non-null  object
 11  Orders            11245 non-null  float64
 12  Amount            11245 non-null  float64
dtypes: float64(5), object(8)
memory usage: 1.1+ MB
```

# Data Cleaning

In [4]:
```
#Deleting blank columns
df.drop(['Zipcode'], axis=1, inplace=True)
```

In [5]:
```
#List of columns available
df.columns
```

Out[5]:
```
Index(['User_ID', 'Cust_name', 'Product_ID', 'Age', 'Age Group', 'Gender',
       'State', 'Zone', 'Profession', 'Product_Category', 'Orders', 'Amount'],
      dtype='object')
```

In [6]:
```
#check for null values
pd.isnull(df).sum()
```

Out[6]:
```
User_ID             12
Cust_name           12
Product_ID          12
Age                 12
Age Group           12
Gender              12
State               12
Zone                12
Profession          12
Product_Category    12
Orders              12
Amount              12
dtype: int64
```

In [9]:
```
#drop null values
df.dropna(how='all' ,inplace=True)
```

In [10]:
```
df.shape
```

Out[10]:
```
(11245, 12)
```

In [21]:
```
#replace value of Gender column
df['Gender']=df['Gender'].replace('M','Male')
```

In [22]:
```
#View only Male Gender data
df[df['Gender'] =='Male']
```

Out[22]:

| | User_ID | Cust_name | Product_ID | Age | Age Group | Gender | State | Zone |
|---|---|---|---|---|---|---|---|---|
| **3** | 1001425.0 | Virendra | P00237842 | 16.0 | 0-17 | Male | Karnataka | South |
| **4** | 1000588.0 | Vishal | P00057942 | 28.0 | 26-35 | Male | Gujarat | West |
| **5** | 1000588.0 | Suuraj | P00057942 | 28.0 | 26-35 | Male | Himachal Pradesh | Northern |
| **8** | 1003224.0 | Kushal | P00205642 | 35.0 | 26-35 | Male | Uttar Pradesh | Central |
| **11** | 1003829.0 | Harsh | P00200842 | 34.0 | 26-35 | Male | Delhi | Central |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... |
| **11249** | 1005446.0 | Sheetal | P00297742 | 53.0 | 51-55 | Male | Gujarat | West |
| **11250** | 1005446.0 | Sheetal | P00297742 | 53.0 | 51-55 | Male | Madhya Pradesh | Central |
| **11252** | 1000695.0 | Manning | P00296942 | 19.0 | 18-25 | Male | Maharashtra | West |
| **11253** | 1004089.0 | Reichenbach | P00171342 | 33.0 | 26-35 | Male | Haryana | Northern |
| **11255** | 1004023.0 | Noonan | P00059442 | 37.0 | 36-45 | Male | Karnataka | South |

3408 rows × 12 columns

# Let's Learn EDA - Exploratory Data Analysis

In [23]:

```
# Use describe() method to return description of data in the DataFrame.
df.describe()
```

Out[23]:

| | User_ID | Age | Orders | Amount |
|---|---|---|---|---|
| **count** | 1.124500e+04 | 11245.000000 | 11245.000000 | 11245.000000 |
| **mean** | 1.003004e+06 | 35.415651 | 3.500311 | 9461.934237 |
| **std** | 1.716207e+03 | 12.756369 | 1.713706 | 5234.426634 |
| **min** | 1.000001e+06 | 12.000000 | 1.000000 | 188.000000 |
| **25%** | 1.001492e+06 | 27.000000 | 2.000000 | 5443.000000 |
| **50%** | 1.003065e+06 | 33.000000 | 4.000000 | 8109.000000 |
| **75%** | 1.004429e+06 | 43.000000 | 5.000000 | 12683.000000 |
| **max** | 1.006040e+06 | 92.000000 | 6.000000 | 29350.000000 |

In [24]:

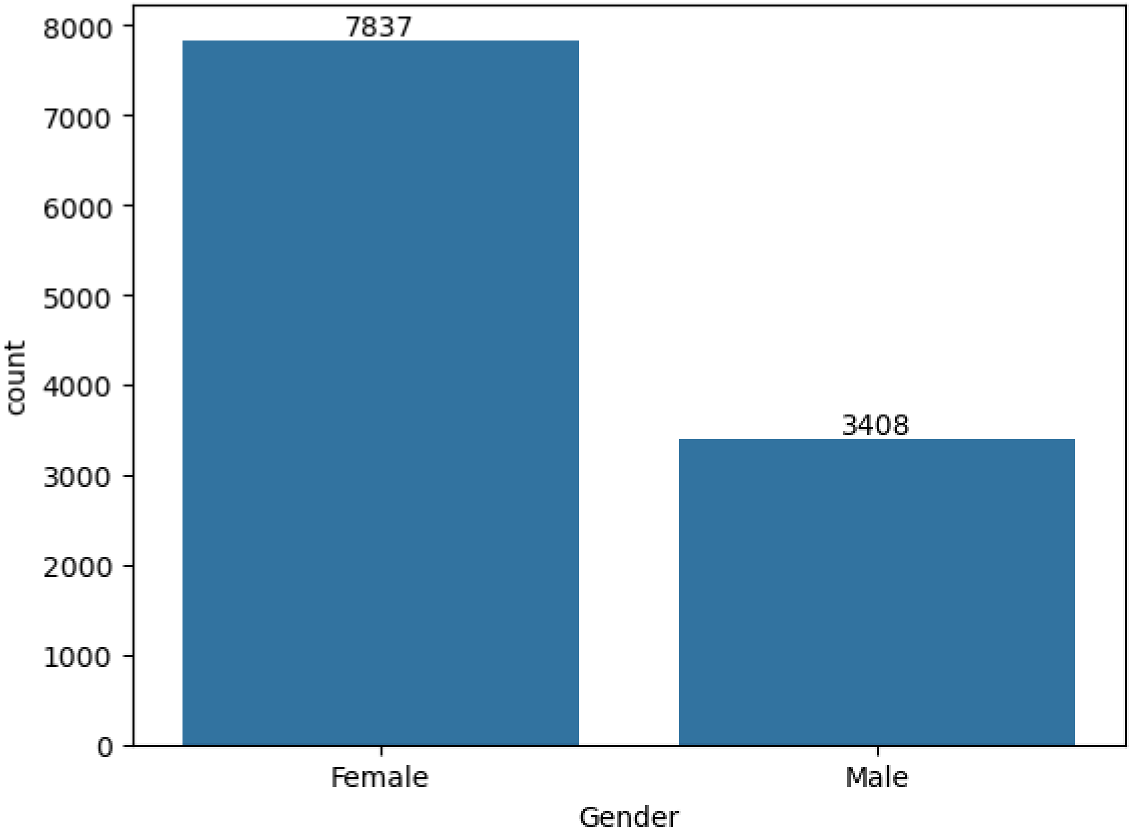
```
#Use describe for specific columns
df[['Age', 'Orders', 'Amount']].describe()
```

Out[24]:

| | Age | Orders | Amount |
|---|---|---|---|
| **count** | 11245.000000 | 11245.000000 | 11245.000000 |
| **mean** | 35.415651 | 3.500311 | 9461.934237 |
| **std** | 12.756369 | 1.713706 | 5234.426634 |
| **min** | 12.000000 | 1.000000 | 188.000000 |
| **25%** | 27.000000 | 2.000000 | 5443.000000 |
| **50%** | 33.000000 | 4.000000 | 8109.000000 |
| **75%** | 43.000000 | 5.000000 | 12683.000000 |
| **max** | 92.000000 | 6.000000 | 29350.000000 |

In [26]:

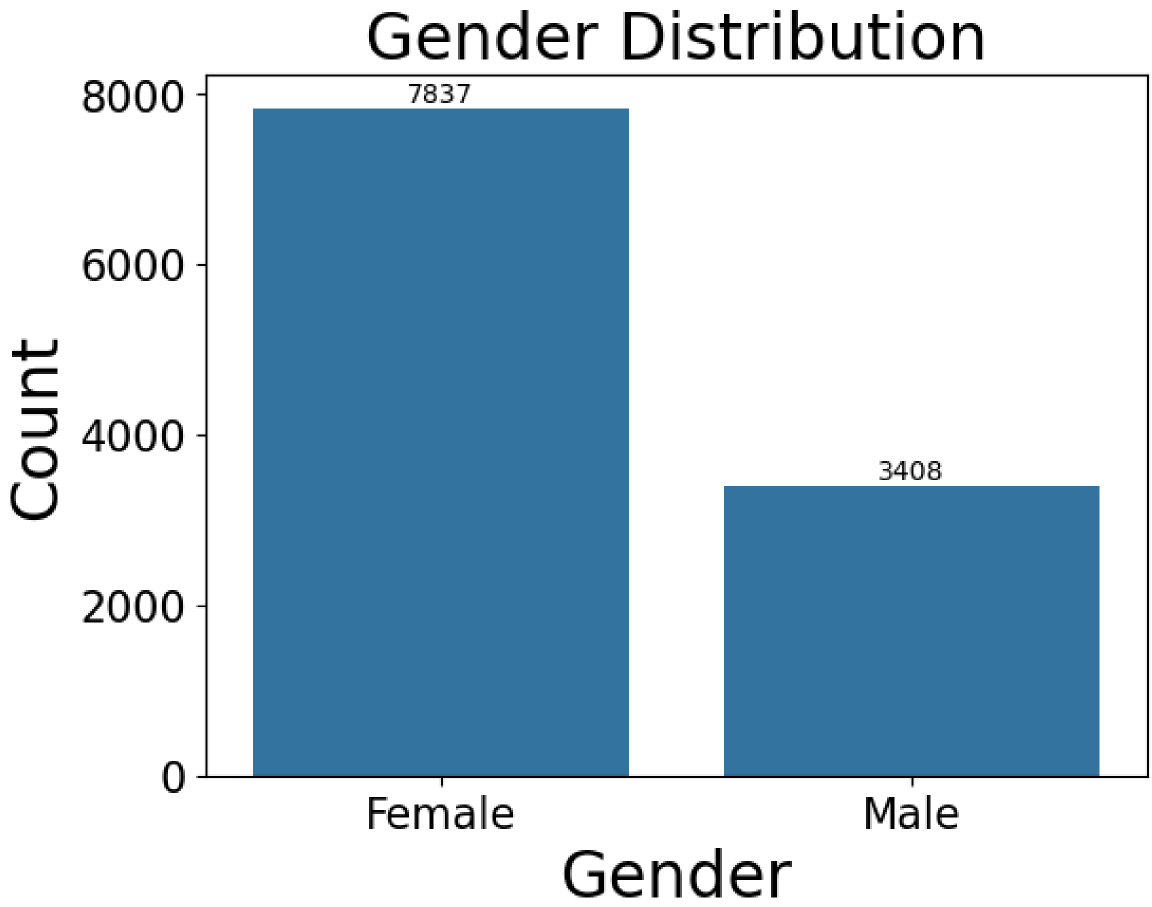
```
ax=sns.countplot(x='Gender' , data= df)
for bars in ax.containers:
    ax.bar_label(bars)
plt.show()
```




In [29]:

```
#Total Transactions count by Gender wise in bar chart
ax=sns.countplot(x= 'Gender', data=df)


#Set Tital and labels with font size
for bars in ax.containers:
    ax.bar_label(bars)
    ax.set_title('Gender Distribution',fontsize=24)
    ax.set_xlabel('Gender', fontsize=24)
    ax.set_ylabel('Count', fontsize=24)
```

```
    ax.tick_params(axis='both', labelsize=16)
plt.show()
```

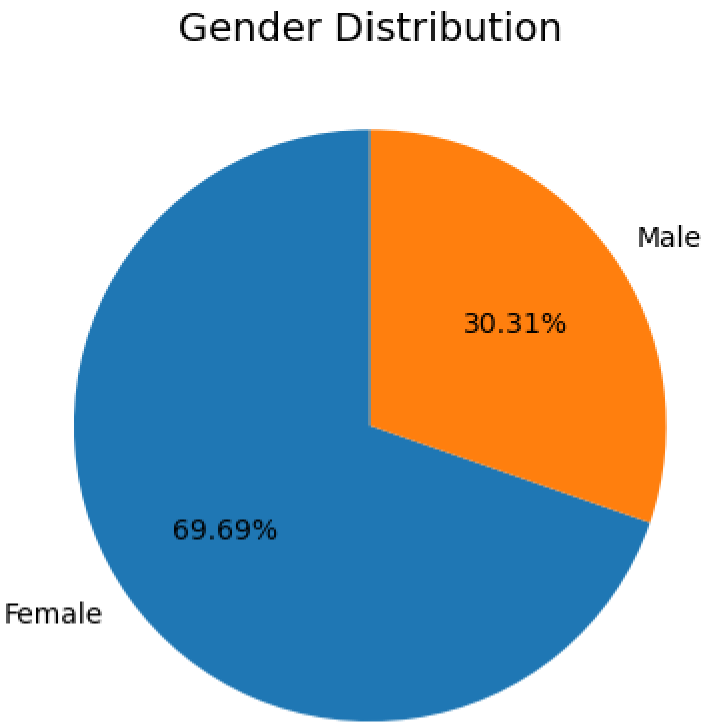



In [31]:
```
# 2. Gender wise distribution in pie chart
```

In [33]:
```
# Total Transaction Count by Gender wise in Pie chart
gender_counts= df['Gender'].value_counts()
print(gender_counts)
```

```
Gender
Female    7837
Male      3408
Name: count, dtype: int64
```

In [35]:
```
#Total Transactions Count by Gender wise in Pie Chart
gender_counts= df['Gender'].value_counts()

plt.pie(
    gender_counts,
    labels=gender_counts.index,
    autopct='%.2f%%',
    startangle=90,
)

#Add a title
plt.title('Gender Distribution', fontsize=14)
plt.show()
```




In [36]:
```
df.groupby('Gender', as_index=False)['Amount'].sum()
```

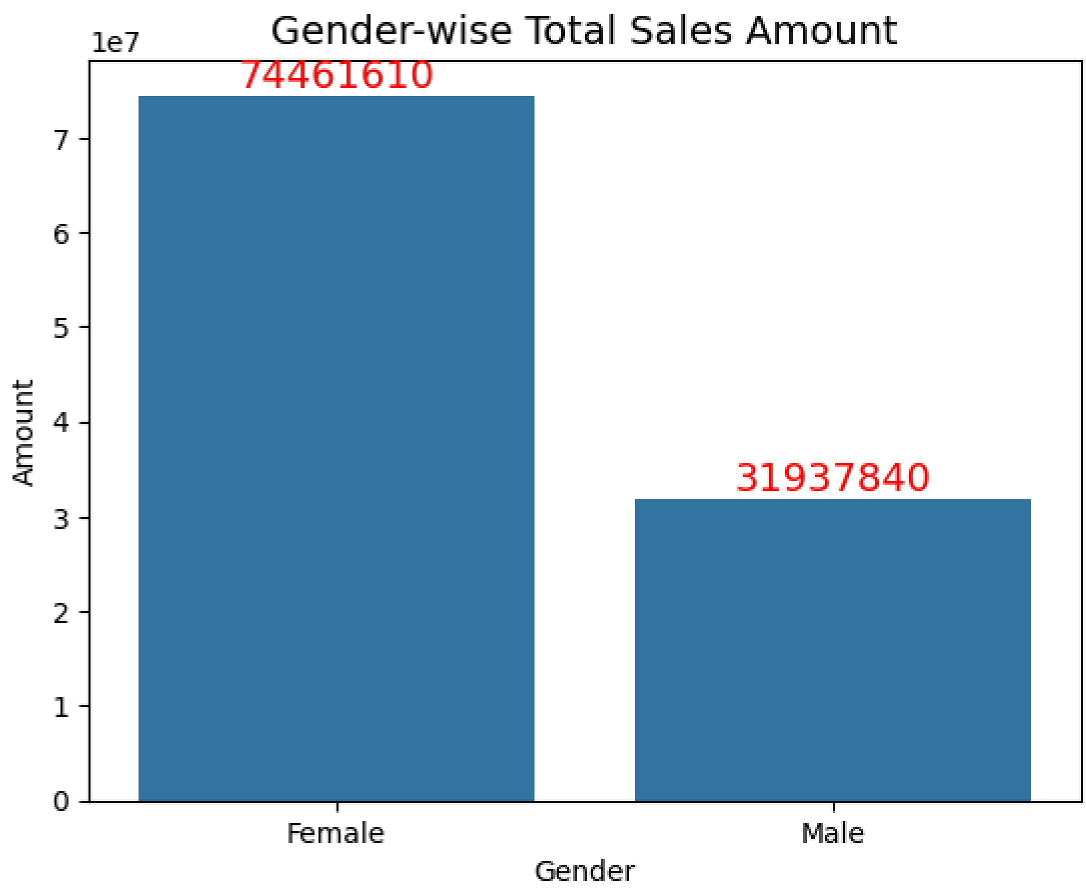
Out[36]:

| | Gender | Amount |
|---|---|---|
| **0** | Female | 74461610.49 |
| **1** | Male | 31937840.00 |

In [37]:
```
Gen_Wise_Sales= df.groupby('Gender', as_index=False)['Amount'].sum().sort_values
print(Gen_Wise_Sales)
```

```
   Gender       Amount
0  Female  74461610.49
1    Male  31937840.00
```

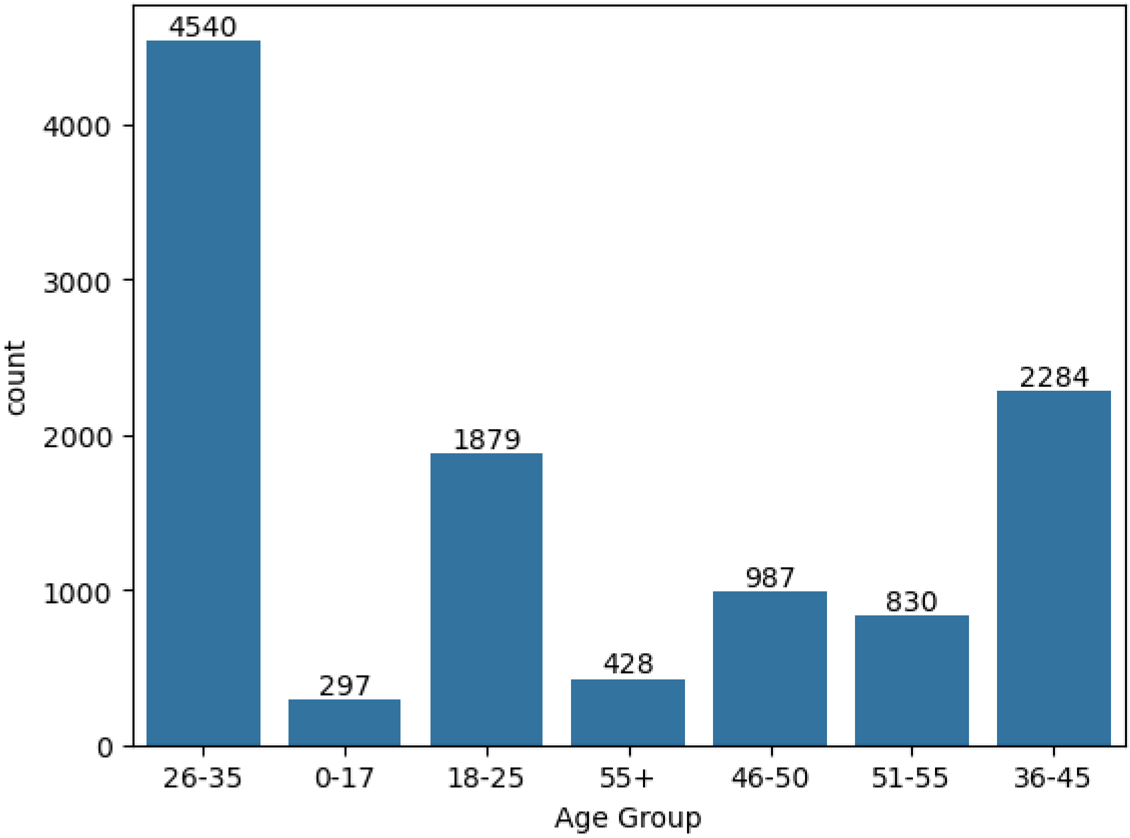
In [44]:
```
#Gender wise Total Sales Amount

#Step 1: Group data by Gender and Sum the amount
Gen_Wise_Sales=df.groupby('Gender', as_index=False)['Amount'].sum().sort_values(

#Step 2: Create the bar plot
ax=sns.barplot(x='Gender', y='Amount', data=Gen_Wise_Sales)

# Step 3: Add data lables on topn of bars
for bar in ax.containers:
    ax.bar_label(bar, fmt='%.0f', fontsize=14, color='red')

# Step 4: Add titles
plt.title('Gender-wise Total Sales Amount', fontsize=14)
plt.show()
```




In [ ]:
```
Conclusion : Male sales are higher than female sales
```

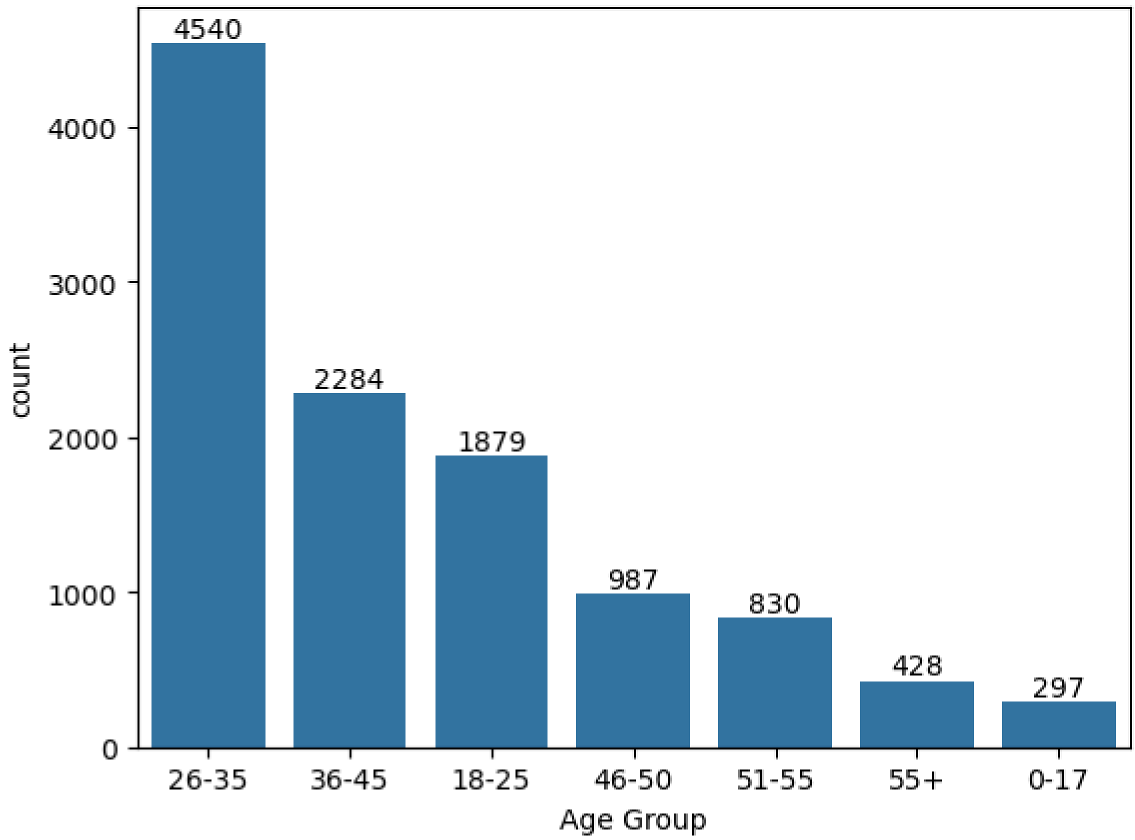
# -->Age group wise total transactions

In [46]:
```
# Age Group wise Transactions Count in Bar Chart
ax= sns.countplot(data= df, x='Age Group')

for bars in ax.containers:
    ax.bar_label(bars)
plt.show()
```

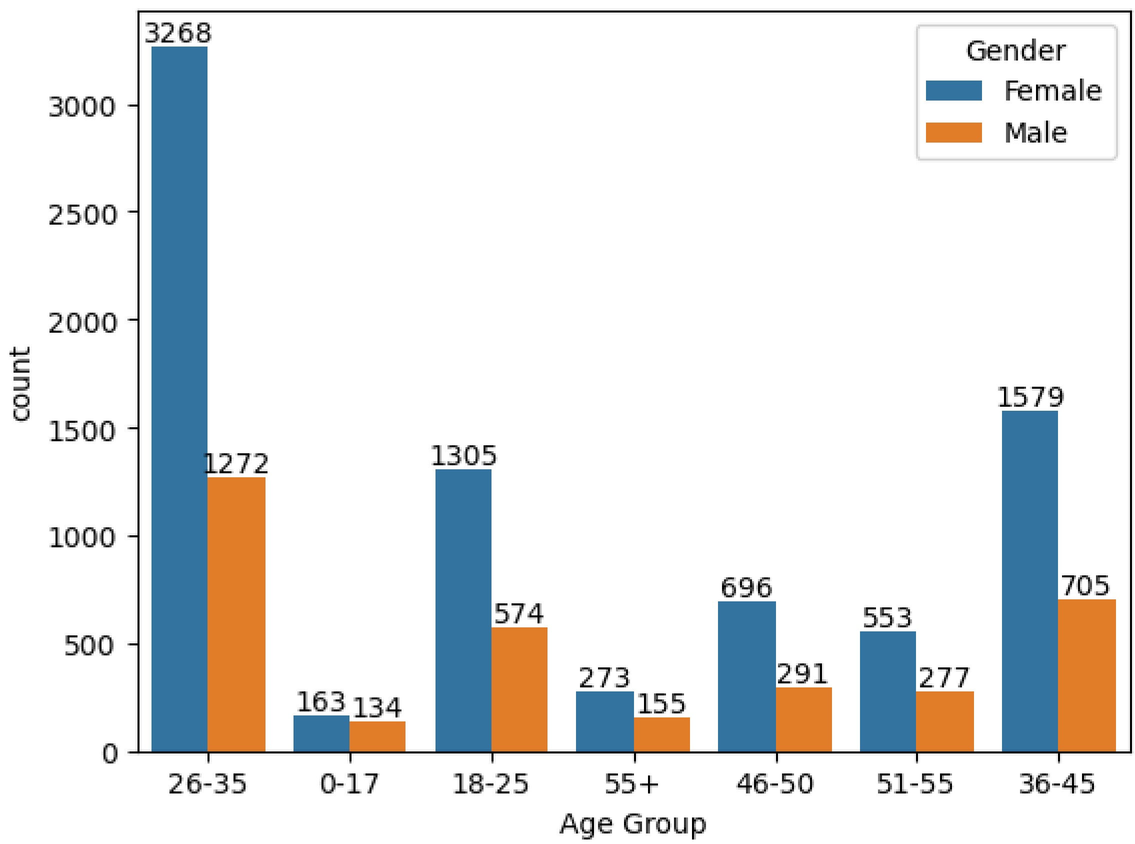



In [48]:

```
# Step1 -Age group wise transactions count (Sorting)

age_group_counts=df['Age Group'].value_counts().sort_values(ascending=False)

# Step2 -Use Ordered categories for sorting in countplot
sns_order=age_group_counts.index

#Age Group wise Transactions Count in Bar Chart
ax=sns.countplot(data=df, x='Age Group', order=sns_order)

for bars in ax.containers:
    ax.bar_label(bars)
plt.show()
```

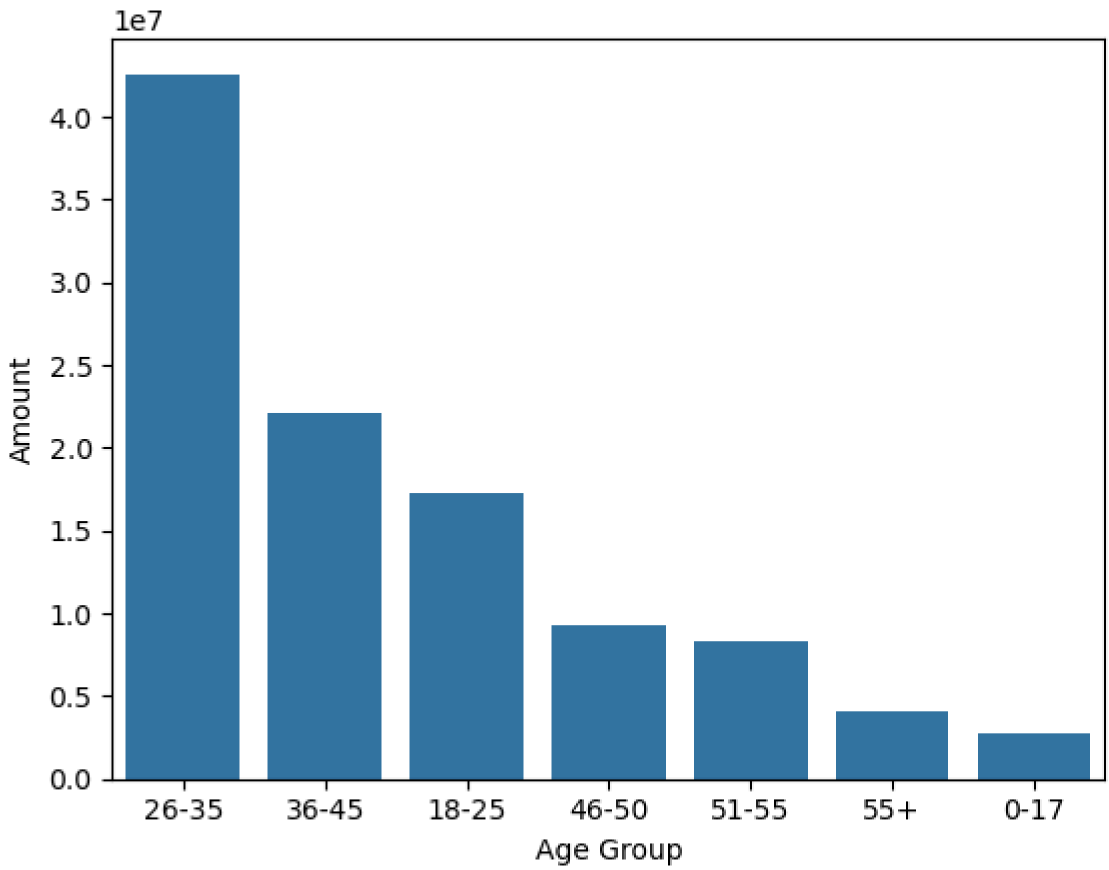



In [49]:
```
# Age Group and Gender wise Transactions distribution

ax=sns.countplot(data=df, x='Age Group', hue='Gender')

for bars in ax.containers:
    ax.bar_label(bars)
plt.show()
```



In [ ]:
```
From above graphs we can see that most of the buyers are of age group between 26
```

In [51]:
```
# Age Group wise Total Amount
sales_age=df.groupby(['Age Group'], as_index=False)['Amount'].sum().sort_values(

sns.barplot(x= 'Age Group', y='Amount', data = sales_age)

plt.show()
```



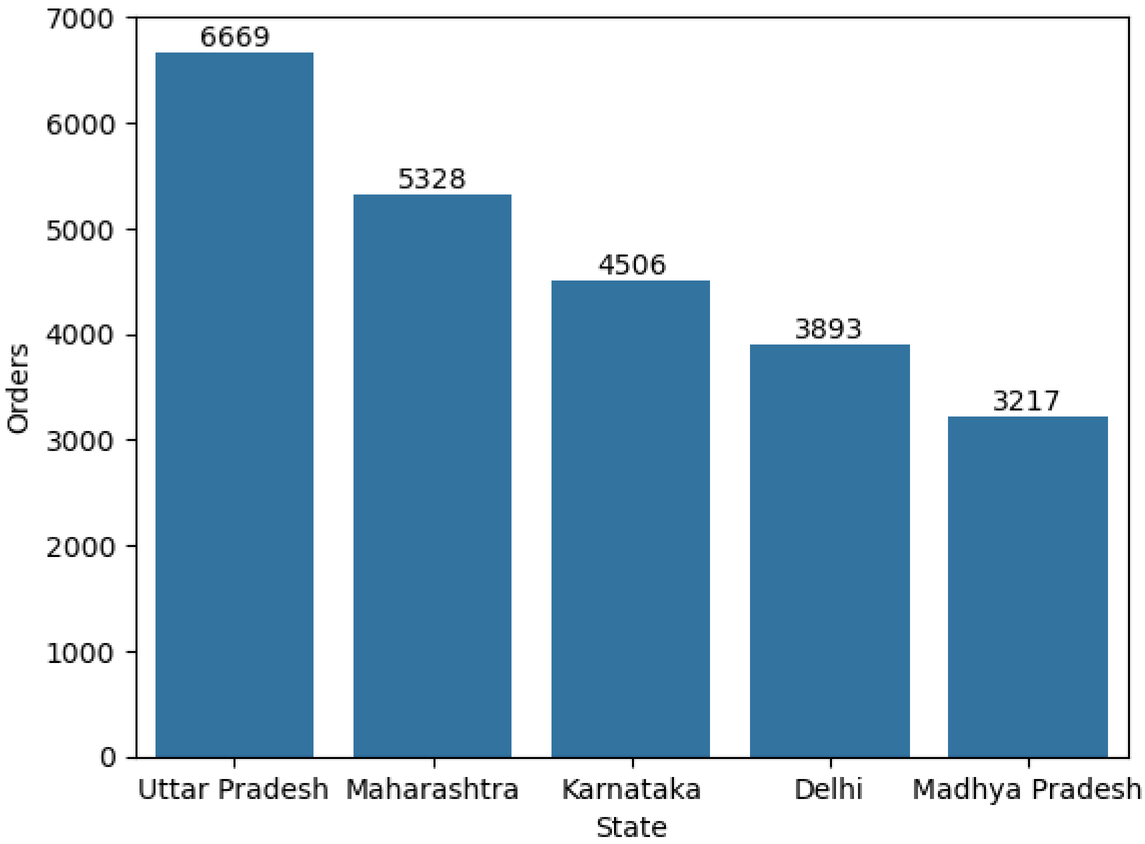

# --> State wise analysis

In [4]:
```
order_state= df.groupby(['State'], as_index=False)['Orders'].sum().sort_values(b
print(order_state)
```

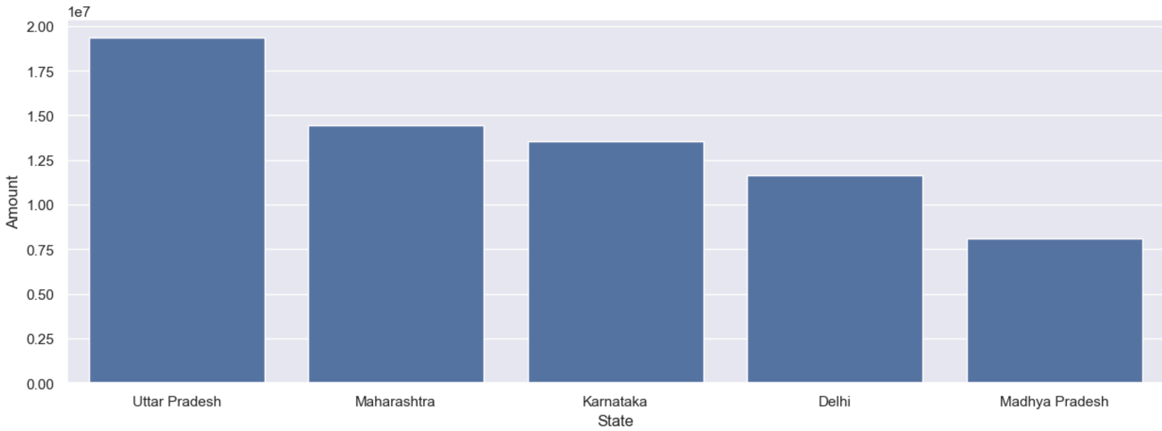
```
             State  Orders
14   Uttar Pradesh  6669.0
10     Maharashtra  5328.0
7        Karnataka  4506.0
2            Delhi  3893.0
9   Madhya Pradesh  3217.0
```

In [5]:
```
# Order wise Top 5 state
order_state= df.groupby(['State'], as_index=False)['Orders'].sum().sort_values(b

#sns.set(rc={'figure.figsize':(15,5)})
ax=sns.barplot(data=order_state, x='State', y='Orders')

for bars in ax.containers:
    ax.bar_label(bars)
```

```
plt.show()
```




In [8]:
```
#Amount Wise top 5 state

sales_state=df.groupby(['State'], as_index=False)['Amount'].sum().sort_values(by

sns.set(rc={'figure.figsize':(15,5)})
sns.barplot(data= sales_state, x='State', y='Amount')
plt.show()
```

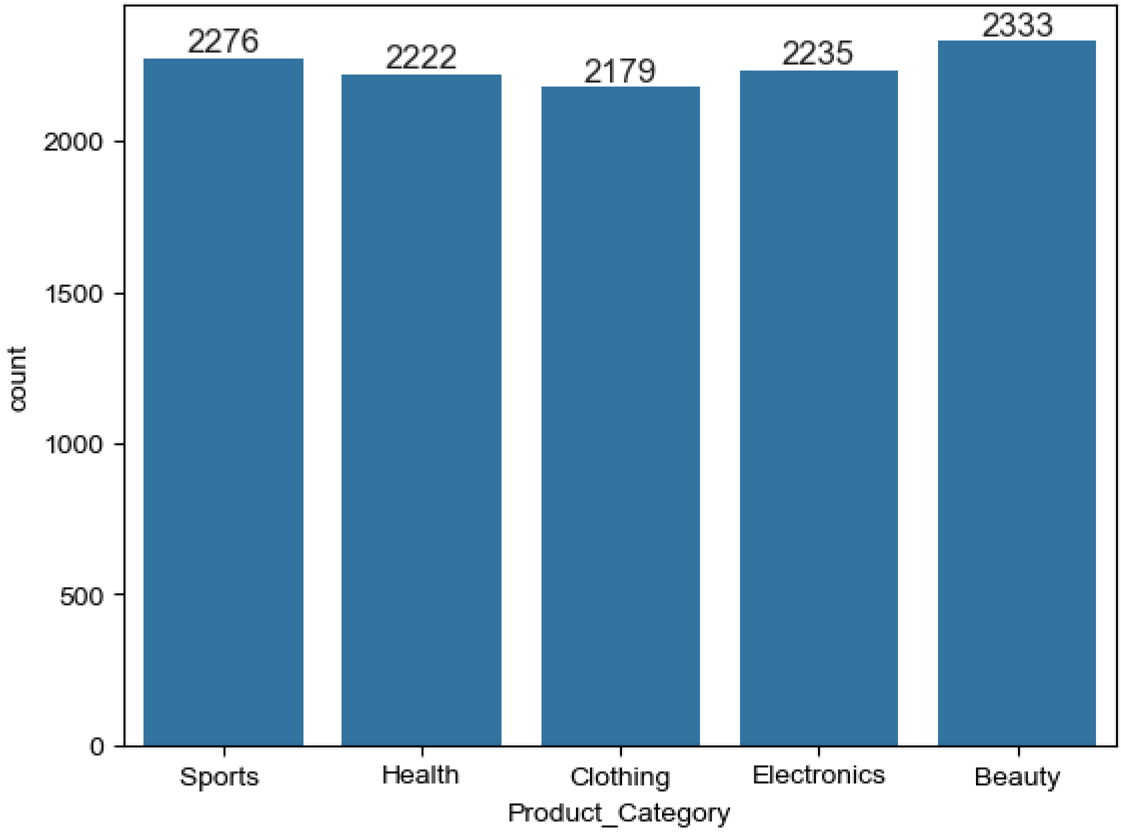



In [ ]:
```
From above graphs we can see that most of the orders & total sales/amount are fr
```

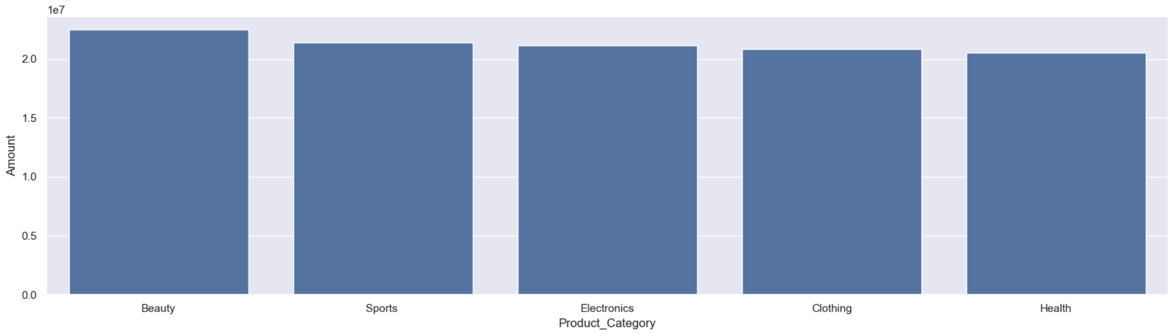
# --> Product_Category Analysis

In [4]:
```
# Product category wise Transactions count
ax=sns.countplot(data=df, x='Product_Category')
```

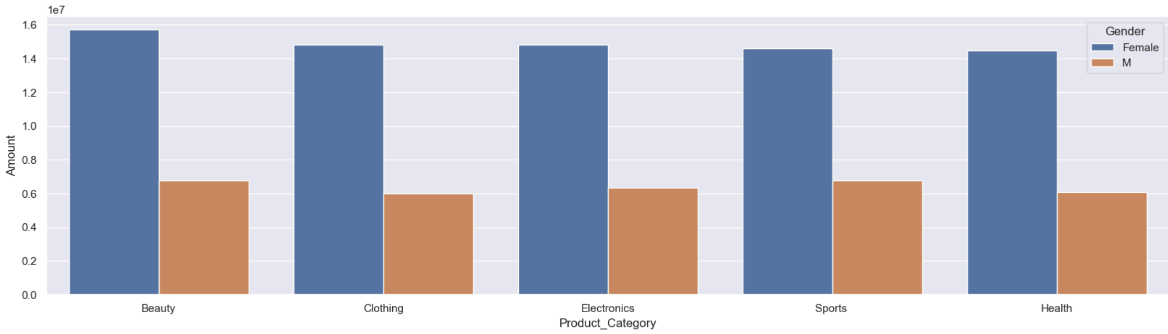
```
sns.set(rc={'figure.figsize':(7,5)})
for bars in ax.containers:
    ax.bar_label(bars)
plt.show()
```




In [5]:
```
#Amount wise Product Category in bar chart
sales_state=df.groupby(['Product_Category'], as_index=False)['Amount'].sum().sor

sns.set(rc={'figure.figsize':(20,5)})
sns.barplot(data= sales_state, x='Product_Category',y='Amount')

plt.show()
```



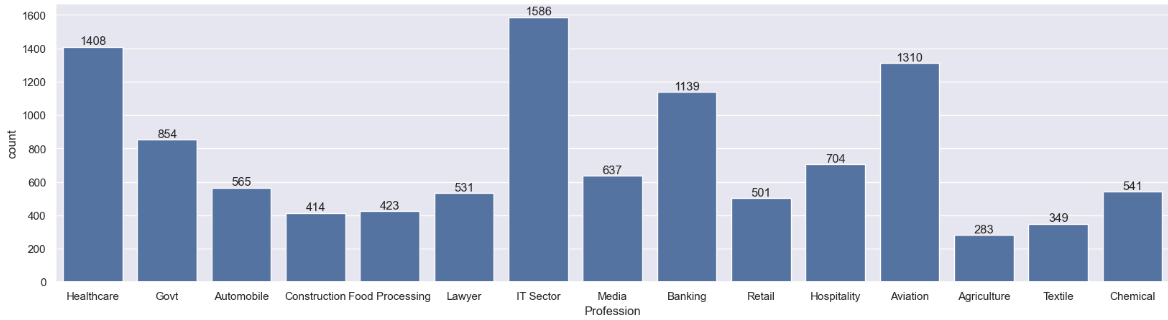

In [10]:
```
#Product category and Gender wise Transactions count
sales_pro = df.groupby(['Product_Category', 'Gender'], as_index=False)['Amount']
sns.barplot(data = sales_pro, x = 'Product_Category',y= 'Amount', hue='Gender')

plt.show()
```



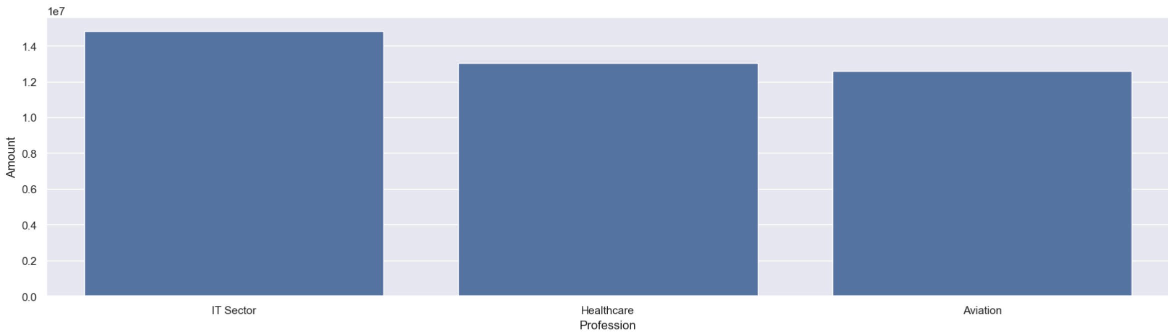

In [ ]:
```
From above graphs we can see that most of the buyers are Female
```

# --> Profession wise Transactions count

In [13]:
```python
# Profession wise Transactions count

ax = sns.countplot(data = df, x = 'Profession')
for bars in ax.containers:
    ax.bar_label(bars)
plt.show()
```



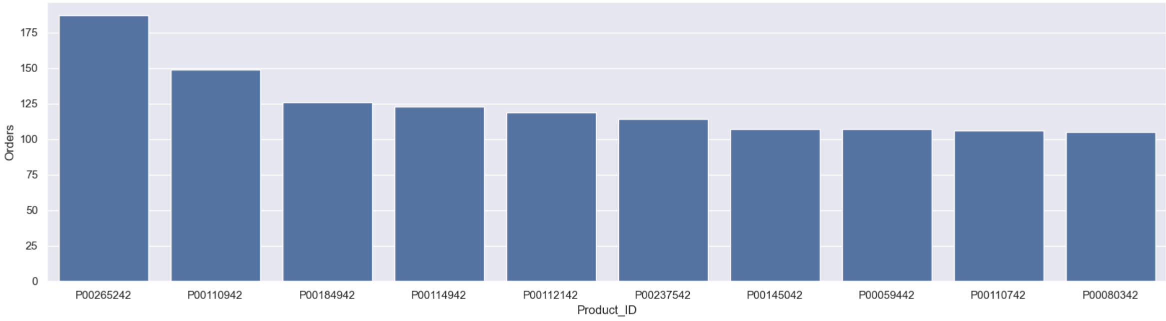

In [15]:
```python
# Amount Wise Top Professions
sales_state = df.groupby(['Profession'], as_index=False)['Amount'].sum().sort_va

sns.set(rc={'figure.figsize':(20,5)})
sns.barplot(data = sales_state, x = 'Profession',y= 'Amount')
plt.show()
```




In [ ]:
```
From above graphs we can see that most of the buyers are working in IT, Healthca
```

In [18]:
```python
#Order wise Top 10 Product
sales_state = df.groupby(['Product_ID'], as_index=False)['Orders'].sum().sort_va

sns.set(rc={'figure.figsize':(20,5)})
```

```
sns.barplot(data = sales_state, x = 'Product_ID',y= 'Orders')
plt.show()
```




In [ ]:
```
Conclusion:
Female age group 26-35 yrs from UP, Maharastra and Karnataka working in IT, Heal
```

In [ ]: